CSE 715

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Quiz-1

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Question - 1 : Task-1 :

Computing mean for each column

$$\mu_B^{(1)} = \frac{2+1+4+5}{4} = \frac{12}{4} = 3$$

$$\mu_B^{(2)} = \frac{4+3+6+7}{4} = \frac{20}{4} = 5$$

$$\mu_B^{(3)} = \frac{6+5+8+9}{4} = \frac{28}{4} = 7$$

$$\therefore \mu_B = [3, 5, 7]$$

Question-2 : Task-2 :

Computing the batch variance vector

$$\sigma_B^2 = \left[\sigma_B^{2(1)}, \sigma_B^{2(2)}, \sigma_B^{2(3)}\right]$$

We know,

Batch variance $\sigma_B^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_B)^2$

$\therefore$ Colum 1 $\sigma_B^{2(1)} = \frac{(2-3)^2 + (1-3)^2 + (4-3)^2 + (5-3)^2}{4}$

$$= \frac{1+4+1+4}{4} = \frac{10}{4} = 2.5$$

Column 2 $\sigma_B^2(2) = \frac{(4-5)^2 + (3-5)^2 + (6-5)^2 + (7-5)^2}{4}$

$= \frac{1+4+1+4}{4} = \frac{10}{4} = 2.5$

Column 3 $\sigma_B^2(3) = \frac{(6-7)^2 + (5-7)^2 + (8-7)^2 + (9-7)^2}{4}$

$= \frac{1+4+1+4}{4} = \frac{10}{4} = 2.5$

$\therefore \sigma_B^2 = [2.5, 2.5, 2.5]$

Task3 :

we know, Normalize $\hat{x_i} = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}$

here $\varepsilon = 10^{-5}$, which is extremely small

$\sigma_B^2 = 2.5 \quad \therefore \sqrt{\sigma_B^2 + \varepsilon} = \sqrt{2.5 + 10^{-5}} \approx 1.5811$

$$x = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 4 & 6 & 8 \\ 5 & 7 & 9 \end{bmatrix}$$

for Columns1:

$\hat{x}_{11} = \frac{x_i - \mu_B^{(1)}}{1.5811}$

$= \frac{2-3}{1.5811} = \frac{-1}{1.5811} = -0.6325$

$\hat{x}_{21} = \frac{x_i - \mu_B^{(1)}}{1.5811}$

$= \frac{1-3}{1.5811} = -1.2649$

$$\hat{x}_{31} = \frac{x_i - \mu_B^{(1)}}{1.5811} = \frac{4-3}{1.5811} \approx 0.6325$$

$$\hat{x}_{41} = \frac{x_i - \mu_B^{(1)}}{1.5811} = \frac{5-3}{1.5811} \approx 1.2649$$

for Column 2:

$$\hat{x}_{12} = \frac{x_i - \mu_B^{(2)}}{1.5811} = \frac{4-5}{1.5811} \approx -0.6325$$

$$\hat{x}_{22} = \frac{x_i - \mu_B^{(2)}}{1.5811} = \frac{3-5}{1.5811} = -1.2649$$

$$\hat{x}_{32} = \frac{6-5}{1.5811} = 0.6325$$

$$\hat{x}_{42} = \frac{7-5}{1.5811} = 1.2649$$

for Column 3:

$$\hat{x}_{13} = \frac{x_i - \mu_B^{(3)}}{1.5811} = \frac{6-7}{1.5811} = -0.6325$$

$$\hat{x}_{23} = \frac{58-7}{1.5811} = -1.2649$$

$$\hat{x}_{33} = \frac{8-7}{1.5811} = 0.6325$$

$$\hat{x}_{43} = \frac{9-7}{1.5811} = 1.2649$$

$$\therefore \hat{x} = \begin{bmatrix} -0.6325 & -0.6325 & -0.6325 \\ -1.2649 & -1.2649 & -1.2649 \\ 0.6325 & 0.6325 & 0.6325 \\ 1.2649 & 1.2649 & 1.2649 \end{bmatrix}$$

Task-4 $\gamma = [1.2, 0.5]$ $\beta = [0, -1, 1]$

We know,

Scale and Shift Output: $Y_i = \gamma \hat{x}_i + \beta$

for Column1: $\gamma_1 = 1, \beta_1 = 0$

$y_{11} = 1.\hat{x}_{11} + 0 = 1, (-0.6325) + 0 = -0.6325$

$y_{21} = 1.x_{21} + 0 = 1. (-1.2649) + 0 = -1.2649$

$y_{31} = 0.6325$

$y_{41} = 1.2649$

for Column 2: $\gamma_2 = 2, \beta_2 = -1$

$\therefore y_{12} = 2.\hat{x}_{12} + (-1) = -1.2649 - 1 = -2.2649$

$y_{22} = 2.\hat{x}_{22} - 1 = 2.5298 - 1 = -3.5298$

$y_{32} = 2.\hat{x}_{32} - 1 = 2(0.6325) - 1 = 1.2649 - 1 = 0.2649$

$y_{42} = 2.\hat{x}_{42} - 1 = 2(1.2649) - 1 = 2.5298 - 1 = 1.5298$

for Column 3: $\gamma = 0.5, \beta_3 = 1$

$\therefore y_{13} = 0.5 \times \hat{x}_{13} + 1 = 0.5(-0.6325) + 1 = 0.6837$

$y_{23} = 0.5 \times \hat{x}_{23} + 1 = 0.5(-1.2649) + 1 = 0.3675$

$y_{33} = 0.5 \times \hat{x}_{33} + 1 = 0.5(0.6325) + 1 = 0.3163 + 1 = 1.3163$

$y_{34} = 0.5 \times \hat{x}_{34} + 1 = 0.5(1.2649) + 1 = 1.6325$

Task-5

Final transformed batch Matrix

$$Y = \begin{bmatrix} -0.6325 & -2.2649 & 0.6837 \\ -1.2649 & -3.5298 & 0.3675 \\ 0.6325 & 0.2649 & 1.3163 \\ 1.2649 & 1.5298 & 1.6325 \end{bmatrix}$$